

Molar volume of an ideal gas and gas laws

Learning outcomes

By the end of this section you should be able to investigate the relationship between temperature, pressure and volume for a fixed mass of an ideal gas and analyse graphs relating these variables.

During hot days, bicycle and car tyres become inflated, increasing the volume of the air inside the tyre. Increasing the temperature causes the particles to move with greater speed, resulting in more frequent collisions with the wall of the tyre, expanding the volume. On colder winter days, the volume of the air inside the tyres decreases. The gas particles then move more slowly, having less frequent collisions with the wall of the tyre. This is often why car or bike tyres need to be inflated or deflated a bit. This is just one example of the relationship between variables describing gases, here between volume and temperature.

Molar volume

Recall from [section](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-hydrogen-emission-spectrum-id-44042/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-hydrogen-emission-spectrum-id-44042/>) S1.5.2 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/real-gases-id-44351/>) that gases behave ideally under low pressure and high temperature, allowing the particles to move freely without influence by the other gas particles. Since temperature and pressure conditions vary from place to place, affecting the behaviour of a gas, scientists often refer to standard conditions to describe the behaviour of gases so that their experiments can be reproduced. One set of standard conditions is called STP – standard temperature and pressure – where the temperature is 273 K (0 °C) and the pressure is 100 kPa.

Under standard conditions of temperature and pressure (STP), the volume of 1 mole of any gas will always occupy 22.7 dm³. This value is called the molar volume (volume of 1 mole) of a gas at STP. This can be used to convert between volume and moles of gas at STP. The formula that can be used is:

$$\text{Amount (in moles)} = \frac{\text{volume (dm}^3\text{)}}{\text{molar volume (dm}^3 \text{ mol}^{-1}\text{)}}$$

or, at STP:

$$\text{Amount (in moles)} = \frac{\text{volume (dm}^3\text{)}}{22.7 \text{ (dm}^3 \text{ mol}^{-1}\text{)}}$$

Study skills

The molar volume of a gas at STP is $22.7 \text{ dm}^3 \text{ mol}^{-1}$. Conditions of STP are a temperature of 273 K and a pressure of 100 kPa. This information can be found in section 4 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/unit-conversions-and-standard-conditions-id-45106/>) of the DP chemistry data booklet.

Be careful not to confuse the conditions of standard temperature and pressure (STP) with those of standard conditions which you will come across in thermochemistry sections. Standard conditions in thermochemistry include a temperature of 298 K.

In section 4 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/unit-conversions-and-standard-conditions-id-45106/>) of the data booklet conditions are given for the SATP, which is the standard ambient temperature and pressure. The temperature at SATP is 298 K rather than 273 K.

Worked example 1

Determine the volume occupied by 0.500 mol of chlorine gas (Cl_2) at STP.

Solution steps	Calculation
Step 1: Write out the values given in the question.	$n = 0.500$ at STP 22.7
Step 2: Write out the equation.	$V =$
Step 3: Substitute the values given.	$V =$
Step 4: State the answer with appropriate units and the number of significant figures used in rounding.	$V =$

Worked example 2

Determine the amount (in mol) of helium gas in a balloon of volume of 33.0 dm^3 at STP.

Solution steps	Calculations
Step 1: Write out the values given in the question.	$V = 33.0 \text{ dm}^3$ at STP, molar v
Step 2: Write out the equation.	$\text{mol } (n) = \frac{\text{---}}{\text{molar volume}}$
Step 3: Substitute the values given.	$n = \frac{33.0}{22.7}$
Step 4: State the answer with appropriate units and the number of significant figures used in rounding.	$n = 1.45 \text{ mol}$

Recall from [section S1.4.6 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/avogadros-law-of-combining-volumes-id-44265/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/avogadros-law-of-combining-volumes-id-44265/), when you learned about Avogadro's law – that equal volumes of gases at the same temperature and pressure contain equal numbers of particles. What happens when we change variables affecting gases? In this section, we will explore relationships between various variables describing gases, namely volume and temperature, volume and pressure, volume and moles, and pressure and temperature. These relationships are described by the gas laws (Boyle's law, Charles' law, and Gay-Lussac's law). The general trends in behaviour with changing conditions of temperature and pressure are the same for all gases. In each case one of these parameters is kept constant, while the relationship between the other two is explored.

Study skills

The names of specific gas laws will not be assessed but you will need to be able to describe the behaviour of a gas when subjected to changes in temperature, volume and pressure.

Relationship between volume and pressure

Boyle's law named after the Irish scientist Robert Boyle (1627–1691), states that at constant temperature the pressure and volume of a fixed mass of an ideal gas are inversely proportional to each other (**Figure 1**). So, if the pressure of a fixed mass of gas at constant temperature is doubled, the volume halves. This can be seen when the plunger of a closed syringe is pushed: as more pressure is exerted, the volume of the enclosed air decreases.

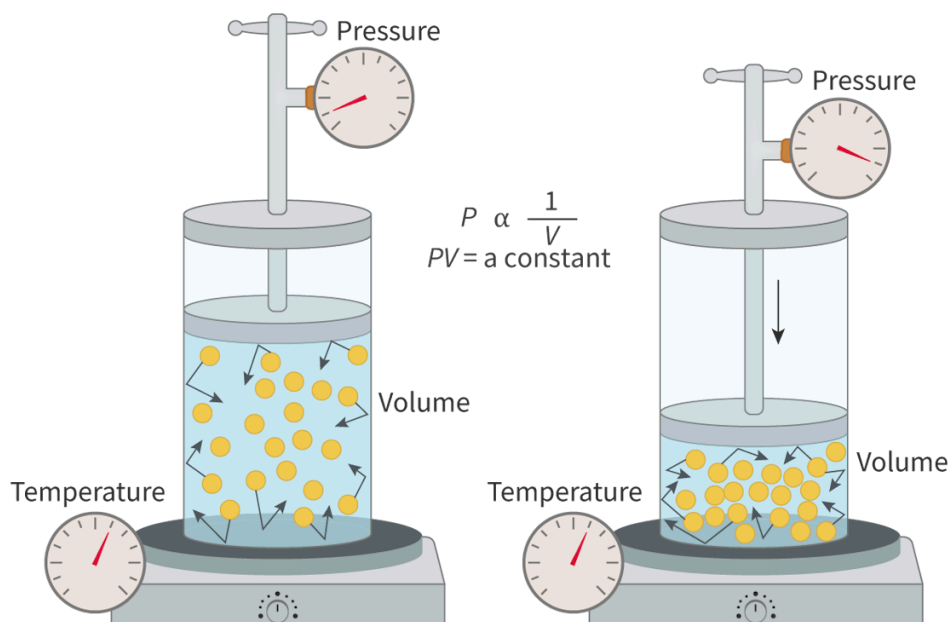


Figure 1. As the pressure on a gas is doubled, its volume is halved.

 More information for figure 1

The image shows two cylindrical containers side by side, each with a movable piston at the top and both under constant temperature. The left container has a lower pressure gauge reading and a larger volume, while the right container has a higher pressure gauge reading and a smaller volume. Yellow particles, representing gas molecules, are inside both containers exhibiting random motion. This demonstrates Boyle's Law, where pressure and volume of a gas are inversely proportional when temperature is constant. Arrows indicate the movement of gas particles. A formula is noted between the containers indicating the inverse relationship: ($P \propto \frac{1}{V}$) and ($PV = \text{a constant}$).

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This relationship can be visualised graphically as in **Figure 2**. Look carefully at the plotted variables below. What do you notice?

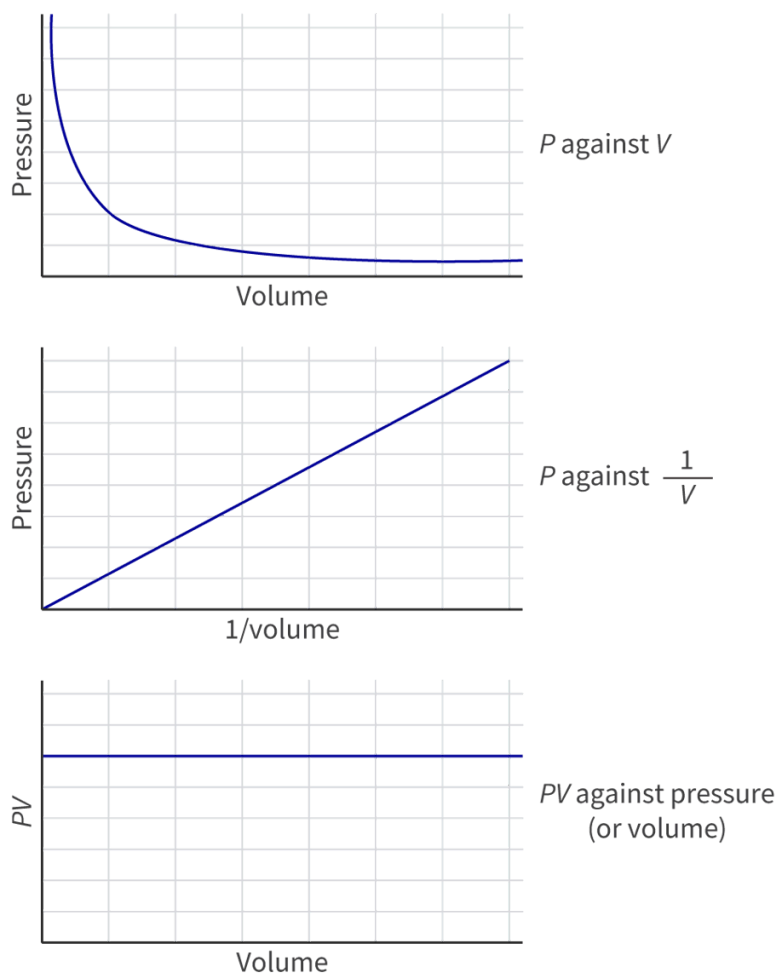


Figure 2. Boyle's law graphical illustration.

 More information for figure 2

The image contains three graphs illustrating Boyle's law relationships.

1. The first graph shows "Pressure" on the y-axis versus "Volume" on the x-axis. The curve illustrates an inverse relationship, where the pressure decreases as the volume increases, depicted by a downward sloping curve.
2. The second graph represents "Pressure" on the y-axis against "1/Volume" on the x-axis. In this graph, a linear positive correlation is shown with a straight line moving diagonally upward, indicating that pressure is directly proportional to the inverse of volume.
3. The third graph displays "P*V" on the y-axis, captioned as "PV against pressure (or volume)". This graph shows a horizontal line, suggesting that the product of pressure and volume remains constant regardless of changes in individual variables.

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As observed in the above graphs, pressure and volume are inversely proportional

$$\left(P \propto \frac{1}{V}\right).$$

As such, Boyle's law can be represented in equation form as:

$$PV = k$$

Therefore, a plot of P against $\frac{1}{V}$ would yield a straight line.

This also means that the product of the pressure and volume of an ideal gas at a certain temperature is constant, and for two conditions of pressure and volume for a sample of gas at constant temperature, we can write the equation shown in

Table 1.

Table 1. Equation for two conditions of pressure and volume for a sample of gas at constant temperature.

Equation	Symbols	Units
$P_1V_1 = P_2V_2$	P_1 = pressure	kiloPascals (kPa)
	P_2 = pressure	kiloPascals (kPa)
	V_1 = volume	cubic decimetres (dm ³)
	V_2 = volume	cubic decimetres (dm ³)

Worked example 3

A 2.20 dm³ gas sample at a pressure of 100. kPa is compressed to 80.0 kPa at constant temperature. Calculate its new volume.

Solution steps

Step 1: Write out the values given in the question.

Step 2: Write out the equation.

Solution steps**Step 3:** Rearrange the equation to make V_2 the subject.**Step 4:** Substitute the values given.**Step 5:** State the answer with appropriate units and the number of significant figures used in rounding.

Relationship between volume and temperature

Charles' law, named after the scientist Jacques Charles, states that at constant pressure the volume of a fixed mass of an ideal gas is directly proportional to its absolute temperature in kelvin ($V \propto T$). This means that if the absolute temperature of a fixed mass of an ideal gas is doubled, the volume of the gas will also double. Charles' law can be represented in equation form as $\frac{V}{T} = k$.

Therefore, the ratio of volume to temperature is constant, and for two conditions of temperature and volume for a sample of gas at constant pressure, we can write the equation shown in **Table 2**.

Table 2. Equation for two conditions of temperature and volume for a sample of gas at constant pressure.

Equation	Symbols	Units
$\frac{V_1}{T_1} = \frac{V_2}{T_2}$	$V_1 = \text{volume}$	cubic decimetres (dm ³)
	$V_2 = \text{volume}$	cubic decimetres (dm ³)
	$T_1 = \text{temperature}$	kelvin (K)
	$T_2 = \text{temperature}$	kelvin (K)

This is what happens in car or bicycle tyres on hot days: as the temperature of the air inside the tyres increases, the volume of the air increases, inflating the tyres!

Worked example 4

A 2.20 dm^3 gas sample at 256 K is cooled down to a temperature of 100 K while keeping its pressure constant. Calculate its new volume.

Solution steps

Step 1: Write out the values given in the question.

Step 2: Write out the equation.

Step 3: Rearrange the equation to make V_2 the subject.

Step 4: Substitute the values given.

Step 5: State the answer with appropriate units and the number of significant figures used in rounding.

So what would a graph of volume vs. temperature yield? What do you expect the graph to look like? Since volume and temperature are directly proportional, then the graph will be a straight line. Charles' law is illustrated graphically in **Figure 3**.

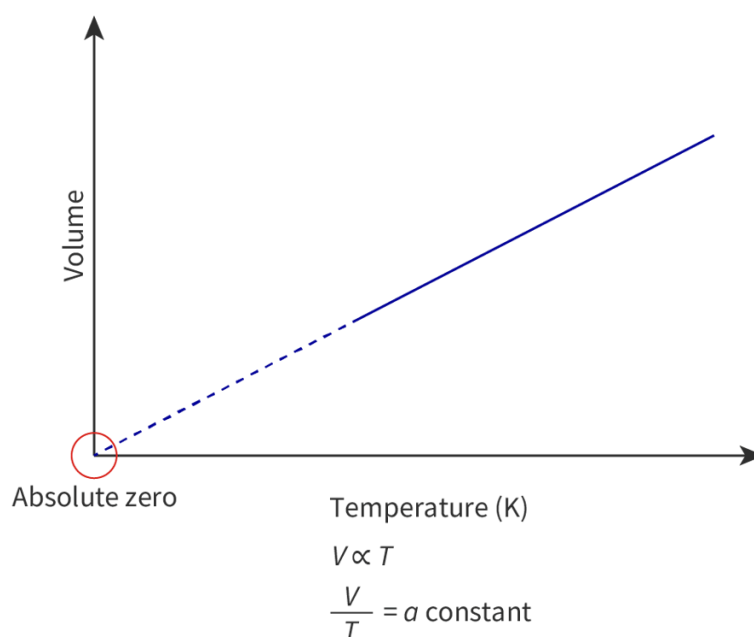


Figure 3. At constant pressure, the volume of a gas is directly proportional to its absolute temperature (in K). The graph can be extrapolated back to absolute zero — the temperature at which a gas would have zero volume.

 More information for figure 3

The image is a graph illustrating the relationship between volume and temperature for a gas. The X-axis represents temperature in Kelvin, ranging from absolute zero upwards. The Y-axis represents volume. This graph demonstrates Charles' Law, showing a direct proportionality between volume and temperature. The graph features a straight, diagonal line starting from the origin (predicting zero volume at absolute zero temperature), extending positively towards the upper right of the graph. Alongside the graph, it has additional text: ' $V \propto T$ ' and ' $V/T = \text{a constant}$ ', which describe the law mathematically. This visual representation highlights the linear relationship described by Charles' Law, indicating that as temperature increases, the volume also increases proportionally, provided the pressure remains constant.

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Worked example 5

A sample of gas occupies 2.20 dm^3 at 30.0°C . Calculate the new volume of this sample when the temperature of this gas is decreased to 15.0°C while keeping its pressure constant.

Solution steps	Calculations
Step 1: Write out the values given in the question.	$V_1 = 2.20 \text{ dm}^3$ $T_1 = 30.0^\circ\text{C}$ $T_2 = 15.0^\circ\text{C}$
Step 2: Write out the equation.	$\frac{V_1}{T_1} = \frac{V_2}{T_2}$
Step 3: Rearrange the equation to make V_2 the subject.	$V_2 = \frac{V_1 T_2}{T_1}$
Step 4: Substitute the values given.	In order to use the formula, they must be in Kelvin (you will get the incorrect answer if you use temperature in $^\circ\text{C}$). $T_1 = 30.0 + 273 = 303 \text{ K}$ $T_2 = 15.0 + 273 = 288 \text{ K}$ $V_2 = \frac{2.20 \times 288}{303} = 2.06 \text{ dm}^3$

Solution steps	Calculations
	$T_1 = 30.0\text{ }^{\circ}\text{C} + 273$ $= 303\text{ K}$ $T_2 = 15.0\text{ }^{\circ}\text{C} + 273$ $= 288\text{ K}$ $V_2 = \frac{(2.20\text{ dm}^3)(288\text{ K})}{(303\text{ K})}$ $V_2 = 2.09\text{ dm}^3$
Step 5: State the answer with appropriate units and the number of significant figures used in rounding.	$V_2 = 2.09\text{ dm}^3$

Relationship between pressure and temperature

Gay-Lussac's law, named after Joseph Louis Gay-Lussac, states that at constant volume the pressure of a fixed mass of an ideal gas is directly proportional to its absolute temperature in kelvin (K). This means that if the absolute temperature of a fixed mass of an ideal gas is doubled, the pressure of the gas will also double.

Gay-Lussac's law can be represented in equation form as $\frac{P}{T} = k$.

Therefore, the ratio of pressure to temperature is constant, and for two conditions of temperature and pressure for a sample of gas at constant volume, we can write the equation shown in **Table 3**. Just as before in Charles' law, the temperature must be expressed in units of kelvin to reflect the absolute temperature of a gas, which would exert no pressure at 0 K, absolute zero.

Table 3. Equation for two conditions of temperature and pressure for a sample of gas at constant volume.

Equation	Symbols	Units
$\frac{P_1}{T_1} = \frac{P_2}{T_2}$	P_1 = pressure	kiloPascals (kPa)
	P_2 = pressure	kiloPascals (kPa)
	T_1 = temperature	kelvin (K)

Equation	Symbols	Units
	$T_1 = \text{temperature}$	kelvin (K)

This relationship can be visualised graphically in **Figure 4**.

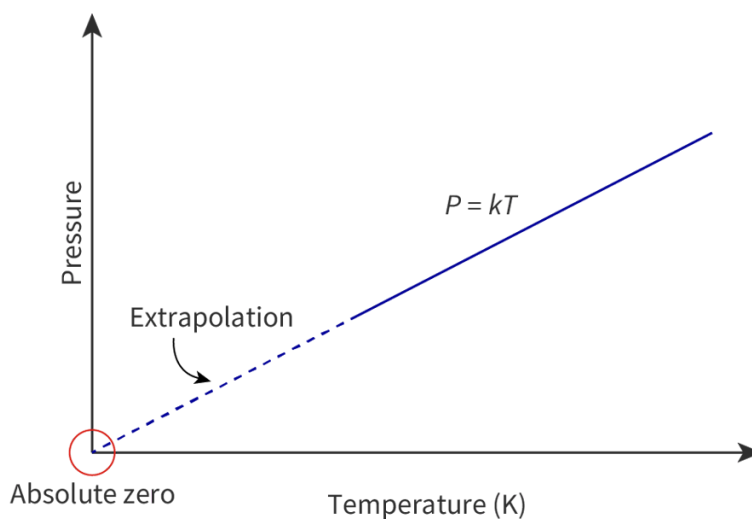


Figure 4. Gay-Lussac's law graphical illustration.

 More information for figure 4

The graph illustrates Gay-Lussac's law, showing the relationship between pressure and temperature. The X-axis represents temperature in Kelvin, starting from absolute zero and extending to higher temperatures. The Y-axis represents pressure, with a linear upward trend indicating a proportional relationship, described by the equation $P = kT$. An extrapolation is indicated, showing the line extending back to theoretically intersect the temperature axis at absolute zero, suggesting that pressure is zero at this point. A solid blue line depicts this relationship with a dashed extension to highlight the extrapolation.

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When a deodorant can is thrown in fire, it explodes! Why? The gas inside heats up, but the volume of the container cannot expand as it is rather rigid. As such, the pressure of the gas increases to an extent that it makes the walls of the can explode.

Worked example 6

A gas sample at 256 K and pressure of 100 kPa is cooled down to a temperature of 100 K while keeping its volume constant. Calculate its new pressure.

Solution steps

Step 1: Write out the values given in the question.

Step 2: Write out the equation.

Step 3: Rearrange the equation to make P_2 the subject.

Step 4: Substitute the values given.

Step 5: State the answer with appropriate units and the number of significant figures used in rounding.

Drag and drop the numbers to make each statement true.

Check your understanding! Drag the correct words into the blanks of the corresponding statements.

The volume of a gas changes by a factor of _____ if the temperature is halved at constant pressure.

The volume of a gas changes by a factor of _____ if the pressure is halved at constant temperature.

The pressure of a gas changes by a factor of _____ if the temperature is quadrupled at constant volume.

The volume of a gas changes by a factor of _____ if both the temperature and pressure are halved.

1 1/2 2 4

✓ Check

Interactive 1. Understanding Gas Behavior: Key Terms.

Activity

- **IB learner profile attribute:** Open-minded
- **Approaches to learning:**
 - Thinking skills
 - Being curious about the natural world
 - Asking questions
 - Applying key ideas and facts in new contexts
 - Social skills — Working collaboratively to achieve a common goal
- **Time required to complete activity:** 20 minutes
- **Activity type:** Pairs activity

Instructions:

In pairs, you will explore relationships between T , P and V .

1. Open the simulation:

<https://teachchemistry.org/classroom-resources/the-gas-laws-simulation> 
(<https://teachchemistry.org/classroom-resources/the-gas-laws-simulation>)n (<https://teachchemistry.org/classroom-resources/the-gas-laws-simulation>)

2. Press on the tab: Boyle's law. Change volume and press calculate, then press on add data. Repeat until you have a clear graph of the relationship. Describe the relationship. What do you notice about temperature?
3. Reset and repeat step two by changing pressure this time.
4. Press on the tab: Charles' law. Change volume and press calculate, then press on add data. Repeat until you have a clear graph of the relationship. Describe the relationship. What do you notice about pressure?
5. Reset and repeat step two by changing the temperature this time.
6. Press on the tab: Gay-Lussac's law. Change pressure and press calculate, then press on add data. Repeat until you have a clear graph of the relationship. Describe the relationship. What do you notice about volume?
7. Reset and repeat step two by changing the temperature this time.